

Green Building Certifications and Property Values in San Francisco

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1. Literature Review

The existing literature on green building certification shares a common core question: Does holding a LEED or Energy Star **label** make a commercial building worth more? Studies differ, however, in how they define "worth more" and in the econometric methods they use to answer that question. Some focus on asking rents, others on effective rents, sale prices, or assessor-reported values. Despite these differences, the findings are consistent: certified buildings command a premium, though the size of that premium varies across studies.

Eichholtz, Kok, and Quigley (2010) provide the most widely cited baseline estimates. Using a national cross-section of U.S. office buildings and a standard hedonic OLS regression, they find that certified buildings earn 3% higher asking rents, over 7% higher effective rents, and approximately 16% higher sale prices. Fuerst and McAllister (2011) ask the same research question, but apply stricter controls by matching certified and uncertified buildings within the same submarket before running the hedonic regression. Their price premium estimates, 25-26%, are larger than those of Eichholtz et al., a difference the authors attribute to better control for location bias in their matched sample. The fact that two studies using similar data sources, but different matching strategies, arrive at different magnitudes highlights that hedonic estimates are sensitive to how the comparison group is constructed. This paper improves on this problem using Coarsened Exact Matching (CEM) on five building characteristics.

Holtermanns and Kok (2017) take a different approach. Rather than a cross-section, they use a panel dataset covering 2004-2013 and ask not just whether certified buildings have higher values today, but whether they appreciate faster over time. Their answer is no: certification raises rent and price levels but does not accelerate rent growth once it becomes common in a market. This is an important nuance that cross-sectional studies cannot detect, and it suggests that the premium is captured at the time of certification rather than compounded, a pattern consistent with what a one-time demand shift in the space market would predict.

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Several studies focus less on the size of the premium and more on explaining why it exists. Devine and Kok (2015) use proprietary billing records and tenant surveys, rather than listing databases, to show that certified buildings achieve 3-4% higher signed rents, 4-10% higher occupancy, lower concessions, and higher tenant satisfaction. This approach directly identifies the operational channels behind the premium: certified buildings perform better, not just market better. Reichardt (2014) isolates one specific channel, operating costs, and finds that LEED buildings have 5.4% lower operating expenses. He finds that the rent premium is only statistically significant under net leases, where tenants directly see and pay utility bills. Under gross leases, where the landlord absorbs operating costs, no significant rent premium appears. This lease-type distinction matters for the present paper because San Francisco's commercial market, this paper's focus, skews toward gross leases, meaning cost savings are more likely to be capitalized into property values than reflected in gross rents. This reinforces the logic of using assessed improvement value as the outcome variable.

Two additional studies round out the literature. Dermisi (2009) is the most directly relevant, as she uses assessor-reported assessed values rather than rents or transaction prices from private databases. She finds that Energy Star is consistently linked to higher assessed values, while LEED effects vary by program and level. Her use of public assessor data confirms that the premium is recognized in administrative valuation systems, not just in private market negotiations, which is the institutional logic behind this paper's data choice. Hui, Chan, and Yu (2015) find a 12.8% rent premium for LEED-certified offices in Shanghai, which is useful for confirming that the value of verified sustainability is not unique to the U.S. institutional context, though differences in leasing customs and regulatory environment limit direct comparability.

Dermisi's approach, however, requires an important assumption: that the assessor income-capitalization method tracks true market value with sufficient accuracy to detect certification-related changes in Net Operating Income (NOI). Sexton, Sheffrin, and O'Sullivan (1995) document that Proposition 13 causes systematic divergence between assessed and market values, with the gap widening for long-held properties that have not had a recent reassessment event. To the extent that assessor capitalization rates lag market conditions, estimated premiums in assessor-based studies, including this one, may be biased in either direction from the true market effect.

Bond and Devine (2016) ask a distinct but related question: Does formal certification outperform unverified green marketing claims? They find that formally certified buildings earn a rent premium nearly double that of buildings making uncertified "green" claims, suggesting it is the third-party verification itself, not just the green narrative, that the market rewards. Von Paumgarten (2003) provides an early qualitative business case for high-performance buildings, arguing that life-cycle cost savings and productivity gains justify higher values. While this study lacks econometric rigor, it anticipated the mechanisms that later empirical work confirmed.

Taken together, the literature has two meaningful gaps that this paper addresses. First, almost all studies use proprietary databases such as CoStar that are inaccessible to most researchers, making replication difficult and long panel designs expensive to construct. This paper uses publicly available San Francisco Assessor-Recorder secured roll data spanning 2007-2024, covering all commercial parcels in the city annually. Second, no prior study directly confronts the renovation confound: certified buildings nearly always undergo major renovations before certification, and it is unclear how much of the observed premium comes from the certification label versus the renovation itself. This paper addresses the confound explicitly through an event study that tests whether pre-certification value trends are parallel, and a renovation-matched comparison that compares certified buildings only against uncertified buildings that also underwent major renovations. This identification strategy is absent from the existing literature and is the primary methodological contribution of this paper.

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2. Conceptual Model

The conceptual model connects sustainability certification to assessed improvement value through two steps: a supply-and-demand model of the office space market, and an income capitalization identity that links space market outcomes to asset values. This two-step structure follows the framework used by Eichholtz et al. (2010) and Fuerst and McAllister (2011), adapted here to explain why the premium should appear in assessor records rather than in rents or transaction prices.

a. The Space Market

In the space market, landlords supply square footage and tenants demand it at a given rent per square foot. For simplicity, I assume the certified and uncertified office markets are segmented: tenants who want certified space compete in one submarket, while the rest compete in another. San Francisco's commercial market is dominated by gross leases, under which landlords, not tenants, bear operating expenses such as utilities, maintenance, and insurance.

A LEED or Energy Star label raises tenant willingness to pay for two reasons: certified space is perceived as higher quality (better air quality, health outcomes, and sustainability credentials), and it signals Environmental, Social, and Governance (ESG) compliance to institutional tenants with sustainability mandates. Both effects shift the demand curve for certified space outward, from D to D' , as shown in Figure 1. Since buildings cannot be certified overnight, the supply of certified space is fixed in the short run, and the supply curve S is vertical. The result is a higher equilibrium rent R^* relative to the uncertified market rent R .

Figure 1. Supply and Demand for Certified Office Space

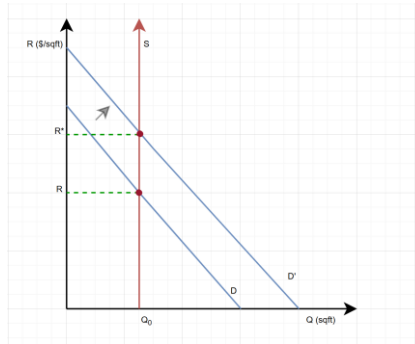


Figure 1. Supply and Demand for Certified Office Space. Certification shifts demand from D to D' , raising equilibrium rent from R to R^* . Supply S is fixed in the short run.

Over time, as more buildings obtain certification, the supply of certified space becomes more elastic and the premium compresses. This is consistent with Holtermanns and Kok (2017), who find that certification raises rent levels but not growth rates once adoption is widespread.

Under gross leases, however, a rent premium is not required for certification to raise property value. Certification-related upgrades, more efficient lighting, HVAC, and building envelope, lower landlord operating expenditure directly. For a given rent level, lower operating expenses raise NOI, and in a capitalized asset market, higher NOI translates into higher property value even if rents remain flat. This cost-side channel is the more consequential mechanism under SF's lease structure and is the primary reason this study uses assessed improvement value, rather than asking rents, as the outcome variable. How the SF Assessor-Recorder translates NOI into assessed improvement value is developed in the Asset Market section below.

b. The Asset Market

Understanding how the SF Assessor-Recorder translates market conditions into an assessed improvement value is essential for interpreting the outcome variable used in the paper. The assessed improvement value is the portion of the total assessed value attributed to the structure itself, excluding land. Land is excluded as its value reflects location, which certification and renovation do not change. Only the improvement value captures changes in building quality and

income-generating capacity. Under California law (California State Board of Equalization, Property Tax Rule 8), assessors must estimate the fair market value of income-producing commercial properties using one of three standard appraisal approaches: the cost approach, the sales comparison approach, or the income capitalization approach. The SF Assessor-Recorder uses the income capitalization approach as the primary method for commercial properties such as offices, hotels, and retail buildings, the types studied here (SF Office of the Assessor-Recorder, 2024). This matters for the research design because certification-related upgrades, like more efficient lighting, HVAC, and building env

elope, lower landlord operating expenses. Under the gross leases that dominate San Francisco's commercial market, operating cost savings flow to the landlord rather than the tenant, directly increasing NOI. A higher NOI, capitalized at the same market rate, produces a higher assessed value. This is what makes assessed improvement value a suitable outcome variable and motivates the research question. Under this approach, the assessed improvement value is calculated as:

$$P = NOI / r \quad (\text{Equation 1})$$

Where P is property value, NOI is net operating income (effective rent * occupancy – operating expenses), and r is the capitalization rate required by investors. Certification raises P by lowering operating expenses, which increases NOI directly and can lower r by reducing the risk of the asset (less depreciation and more stable tenants). For example, an office building in San Francisco with an effective gross income of \$3.5 million and operating expenses of \$1.2 million has an NOI of \$2.3 million. At a market capitalization rate of 5.5%, the assessor would estimate an improvement value of approximately \$41.8 million (\$2.3M ÷ 0.055).

However, under California's Proposition 13 (1978), this income-based calculation is only applied at a reassessment event, typically a change in ownership, a completion of new construction, or issuance of a renovation permit triggering a supplemental assessment. Between reassessment events, assessed values are capped at 2% annual increase regardless of market appreciation. This means a property's assessed value reflects the income approach as applied at its most recent qualifying event, not necessarily current conditions. This institutional feature has an important implication for the renovation subsample used in this study. Buildings that received renovation

permits are more likely to have had a supplemental assessment triggered, meaning their assessed improvement values are more recently updated and better reflect current NOI. The renovation subsample, therefore, serves two purposes. It controls for the confounding effect of concurrent capital investment, and it produces a cleaner measure of certification premium by ensuring assessed values on both sides of the comparison are comparably current.

c. Hypotheses

Both effects above, higher NOI and lower r , predict higher assessed values for certified buildings. Hedonic pricing theory (Rosen, 1974) provides the empirical framework: The assessed value of a building reflects the implicit price of each of its observable characteristics, including its certification status.

The null hypothesis is that certification does not affect assessed improvement values after controlling for building characteristics and market trends ($H_0: \beta_1 = 0$). The alternative, predicted by both the space-market model and the income capitalization identity, is a positive premium ($H_1: \beta_1 > 0$). This prediction is consistent with the premium reported in the prior literature, and specifically with Dermisi (2009), who identifies certification premium in assessor-reported values using a similar hedonic approach. I test these hypotheses using a hedonic cross-section, a difference-in-differences panel, and a renovation-matched design, all of which are developed in the Empirical Framework section below.

3. Data and Methodology

a. Data Sources

This study draws on four primary data sources, which are merged at the parcel level. The main source is the San Francisco Assessor Secured Property Tax Roll, obtained from the city's open data portal (DataSF). I use 18 annual roll files covering fiscal years 2007-08 through 2024-25, each containing parcel-level records of assessed improvement value, square footage, year built, use category, and neighborhood code for all taxable commercial parcels in San Francisco. The outcome variable I am specifically interested in measuring is the assessed improvement value, defined in the Asset Market section above.

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Certification data come from two registries: The U.S. Green Building Council (USGBC) project directory, a private nonprofit registry, and the EPA's Energy Star Portfolio Manager public dataset, a federal registry. Both are filtered to San Francisco commercial buildings with an active certification as of March 2026. Together, the registries identify 415 certified commercial parcels in the study area, 271 LEED only, 312 Energy Star only, and 168 holding both certifications

To link building addresses in the certification registries to parcel identifiers (Assessor Parcel Numbers) in the tax roll, I use San Francisco's Enterprise Addressing System (EAS), a city-maintained address-parcel crosswalk. Finally, building permit records from DataSF are used in the renovation matched sample to classify each parcel by whether it underwent a major renovation (permit types 1, 3, or 6; estimated cost $\geq 100,000$) at any point between 2000 and 2025.

b. Sample Construction and Stylized Facts

Certified buildings are not randomly assigned. They tend to be larger, newer, in higher-demand neighborhoods, and more likely to have recently undergone major renovations. An OLS comparison of certified versus uncertified buildings would therefore conflate the certification premium with pre-existing differences in building quality and location. To address this, I use Coarsened Exact Matching (CEM) to construct a comparison group that is closely comparable to the treated sample on observable dimensions. Unlike propensity score matching, CEM guarantees balance by construction, eliminating model dependence in the matching step. Starting from the 2024-25 assessor roll, I identify all commercially zoned parcels and flag those with a matching LEED or EnergyStar record as treated. I then apply CEM to select a comparison group of uncertified commercial parcels. Matching is performed on four covariates: neighborhood (SF planning district), square footage quintile, number of stories, and construction decade. CEM retains only control parcels that fall in the same coarsened cell as at least one treated parcel, ensuring that treated and control buildings are closely comparable on observable dimensions. After CEM, 69 treated parcels dropped as unmatched, leaving 346 treated and 2,070 control parcels (2,416 total). For the panel analysis, I stack all 18 annual assessor rolls for these parcels, yielding 51,524 parcel-year observations spanning 2007-2024.

The four matching variables, neighborhood, square footage quintile, stories tier, and construction decade, capture the primary drivers of assessed value independent of certification.

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This ensures that treated and control buildings occupy similar positions in the San Francisco commercial market before any certification effect is considered. Within the matched sample, a renovation subsample is further constructed by restricting both treated and control groups to buildings with a recorded major renovation permit; this is the primary identification design and is described in the Specification 3 section below.

The outcome variable is assessed improvement value, the assessor's estimate of the structure's contribution to total parcel value, updated annually via sales evidence, building permit activity, and income-capitalization methods. Because assessed improvement values are highly right-skewed, I apply a natural log transformation. This log specification is standard in hedonic property value research and allows regression coefficients to be interpreted as approximate percentage changes. As shown in Figure 6, the distribution of log assessed improvement values is bimodal and notably shifted rightward for certified buildings (mean = 16.98, SD = 1.86) relative to controls (mean = 11.87, SD = 3.37), indicating that certified buildings are substantially larger and more valuable on average even within the matched sample.

This approach requires an important assumption: that the SF Assessor's income-capitalization methodology translates changes in NOI into assessed improvement value in a timely, accurate manner. In practice, assessor capitalization rates may lag market conditions, and under Proposition 13, values are anchored to historical reassessment events, further attenuating responsiveness to current conditions. The direction of this bias is ambiguous. During periods of cap rate compression, such as San Francisco's commercial market between 2010 and 2019, assessed values likely understate true market values, biasing estimated premiums downward. During periods of cap rate expansion, such as post-2020, the opposite holds. Additionally, certified buildings are more likely to have recent reassessment events due to renovation activity, meaning their assessed values may be more current than those of control buildings, which could inflate the estimated premium. Estimated certification premiums may therefore be biased in either direction, depending on the period and the relative reassessment recency of treated and control buildings

Missing data arises primarily from parcels assessed as land-only or reporting zero improvement values in a given year. In the 2024-25 cross-section, 982 of the 2,416 matched parcels are dropped, yielding a hedonic sample of 1,434 (297 treated, 1,137 control). In the

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panel, year-level coverage ranges from 2,270 to 2,361 unique parcels. These patterns do not appear systematically related to certification status in a way that would bias estimates.

4. Empirical Framework

The Conceptual Model establishes that certification raises assessed improvement values through two channels, reduced operating costs and market signaling, and predicts a positive premium ($H_1: \beta_1 > 0$). I test this prediction using three complementary specifications: a hedonic cross-section, a difference-in-differences panel with two-way fixed effects, and a renovation-matched comparison that constitutes the primary identification design. Each subsequent specification addresses a distinct threat to causal inference.

Specification 1 - Hedonic OLS

The hedonic model estimates the cross-sectional association between certification status and assessed improvement value in the 2024-25 assessor roll, controlling for building and location characteristics:

$$\ln(V_i) = \alpha + \beta_1 \cdot \text{Treated}_i + \beta_2 \cdot \ln(\text{Sqft}_i) + \beta_3 \cdot \text{Stories}_i + \beta_4 \cdot \text{YrBlt}_i + \gamma \cdot \text{Nhood}_i + \delta \cdot \text{UseType}_i + \varepsilon_i \quad (1)$$

where V_i is the assessed improvement value for parcel i , Treated_i is an indicator equal to one if parcel i holds a LEED or EnergyStar certification, $\ln(\text{Sqft}_i)$ is the log of gross square footage, Stories_i is the number of stories, YrBlt_i is the year of original construction, Nhood_i is a vector of neighborhood fixed effects, and UseType_i is a vector of use category indicators (office, retail, hotel, mixed commercial, industrial). CEM sample weights are applied, and heteroskedasticity-robust standard errors (HC1) are used. The coefficient of interest, β_1 , gives the log-point difference in assessed improvement value between certified and uncertified buildings with identical observable characteristics; the implied percentage premium is $(e^{\beta_1} - 1) \times 100\%$. Specification 1 assumes that, after matching and controlling for building characteristics, certified and uncertified buildings are comparable on all dimensions that affect assessed value. This assumption would be violated if higher-quality buildings are simply more likely to seek

certification, in which case the premium reflects quality, not certification. Specification 2 addresses this by using each building as its own control

Specification 2 - Difference-in-Differences Panel

The DiD approach uses variation in when buildings obtained certification across the 18-year panel to remove time-invariant differences between buildings:

$$\ln(V_{it}) = \alpha + \beta_1 \cdot (Treated_i \times Post_{it}) + \lambda_i + \mu_t + \varepsilon_{it} \quad (2)$$

where V_{it} is the assessed improvement value for parcel i in year t , $Treated_i$ is a time-invariant indicator for certified status, $Post_{it}$ equals one in all years at or after parcel i 's first certification year, λ_i are parcel fixed effects that absorb all time-invariant unobserved heterogeneity, and μ_t are year fixed effects that absorb citywide macroeconomic shocks common to all buildings. Standard errors are clustered by parcel to allow for arbitrary serial correlation within buildings over time. Here, β_1 measures the average change in assessed value for certified buildings after certification, relative to what matched uncertified buildings experienced over the same period. The key assumption is parallel trends: that certified and control buildings would have followed the same value trajectory in the absence of certification. I test this using an event-study specification.

$$\ln(V_{it}) = \alpha + \sum_k \delta_k \cdot (Treated_i \times \mathbb{1}[t - cert_i = k]) + \lambda_i + \mu_t + \varepsilon_{it} \quad (3)$$

where k indexes years relative to each building's certification year and δ_k are the year-by-year coefficients (reference period = year -1 , normalized to zero). If parallel trends hold, pre-certification coefficients ($k < 0$) should be indistinguishable from zero. As shown in Figure 4, they are not. Coefficients at $k \leq -3$ and $k = -2$ are -0.331 and -0.160 , both statistically significant, meaning certified buildings were already on a declining value trajectory before certification relative to controls. This is consistent with buildings undergoing renovation before certification, which temporarily depresses assessed values. This pre-trend formally rejects parallel trends for the full sample and motivates the renovation-matched design in Specification 3. Post-certification coefficients are positive and trend upward, consistent with value appreciation following certification

Specification 3 – Renovation-Matched Design

The primary identification strategy of this paper restricts the sample to buildings that underwent major renovation in both the treated and control groups. Using SF building permit records, I classify parcels into four groups: (1) certified and renovated ($N = 284$), (2) renovated

but uncertified (N = 543), (3) certified but not renovated (N = 62), and (4) neither renovated nor certified (N = 1,527). Specifications 3a and 3b compare only groups (1) and (2). In this renovation-balanced subsample, both groups experience construction disruption and reassessment, which controls the renovation confound. If the certification premium persists in this restricted sample, it is more plausibly attributable to the market signal of certification itself rather than renovation-driven appreciation.

Specification 3a - re-estimates the hedonic cross-section from Specification 1 on the renovation subsample.

Specification 3b - re-estimates the DiD panel from Specification 2 on the renovation subsample.

Specification 3c - Permit-Anchored Event Study:

While specifications 3a and 3b establish that a certification premium persists in a renovation balanced sample, they do not address whether the pre-certification negative trend observed in Specification 2's event study is explained by renovation activity. The certification year is a valid anchor only for treated buildings, those that are certified and renovated, so aligning control buildings to a certification date they never receive is artificial. To construct a shared anchor that is real and meaningful for both groups, I use the year of each parcel i 's first major renovation permit:

$$\ln(V_{it}) = \alpha + \sum_k \delta_k \cdot (Treated_i \times 1[t - permit_i = k]) + \lambda_i + \mu_t + \varepsilon_{it} \quad (4)$$

Where $permit_i$ is the fiscal year of parcel i 's first building permit of type 1, 3, or 6 with an estimated cost of at least \$100,000, and k indexes years relative to that permit date (Reference period = year -1). The post period window extends $k = +10$ to accommodate the average gap between permit issuance and certificate receipt, which in this sample is approximately seven years. This length is consistent with the lengthy timeline of major commercial redevelopment projects which often involve planning, entitlement, financing, construction, and stabilization before certification is obtained. If pre-permit coefficients ($k < -1$) are flat and near zero, parallel trends hold from the moment both groups begin renovation activity. A positive and sustained premium emerging around +7, the average certification date would indicate that certification, rather than renovation alone, drives the assessed value gains observed in Specifications 1 and 2.

5. Results

I. Hedonic Cross-Section Estimate (Specification 1)

Table 1, column 1, reports the hedonic OLS regression on the 2024–25 assessor cross-section ($N = 1,434$; 297 treated, 1,137 control). The coefficient on the green certification indicator is 0.419 (robust SE = 0.107, $t = 3.92$, $p < 0.001$), implying a 52.0 percent premium in assessed improvement value. The 95 percent confidence interval spans [0.209, 0.629] and the R^2 is 0.748. This estimate is consistent with the upper range of hedonic premiums in the green-building literature (Eichholtz, Kok, and Quigley 2010; Fuerst and McAllister 2011). Figure 6 shows the full distributional shift in log assessed values between certified and uncertified parcels, and Figure 7 maps the geographic distribution of certified properties across San Francisco's commercial districts.

II. Difference-in-Differences Panel Estimate (Specification 2)

Table 1, column 2, reports the two-way fixed-effects DiD estimate over 51,524 parcel-year observations from 2007 to 2024. The Treated $\times$ Post coefficient is 0.302 (clustered SE = 0.080, $t = 3.79$, $p < 0.001$), implying a 35.2 percent increase in assessed value in the post-certification period relative to matched controls. The smaller magnitude compared to column 1 is expected as parcel fixed effects absorb time-invariant quality differences, so the DiD identifies only the within-building change attributable to certification. Figure 2 shows a line chart of the mean log assessed improvement value by calendar year, 2007–2024, for treated and control groups. It shows the persistent level difference between certified and uncertified buildings and their broad parallel movement in assessed value trends across the 18-year panel.

The event study version (Figure 1) reveals a pre-trend concern. Coefficients at $k \leq -3$ and $k = -2$ are -0.331 and -0.160 , both statistically significant, formally rejecting parallel trends for the full sample. This pattern is consistent with assessed values declining during pre-certification renovation, motivating the renovation-matched design in Specification 3 and the permit-anchored event study in Specification 3c (Figure 9).

III. Renovation-Matched Design (Specification 3a & 3b)

The primary identification strategy of this paper restricts the sample to buildings with a major permit (type 1, 3, or 6; cost $\geq \$100,000$), comparing certified-and-renovated parcels against renovated but uncertified parcels. After dropping parcels with missing or zero

improvement values, the estimation sample is 248 treated and 508 control parcels. Both groups undergo construction and reassessment, so any remaining premium can be attributed to certification rather than renovation. Figure 3 plots raw value trends for this subsample, and Figure 5 summarizes all specification estimates on a common axis for direct comparison. Figure 3 is particularly compelling because it requires no regression machinery to interpret. Simply comparing the raw values of certified-and-renovated buildings against renovated-but-uncertified buildings, the two groups track each other closely before renovation and then diverge steadily in the years that follow, with the certified group pulling ahead. This is the clearest visual evidence in the paper that certification, not construction activity, drives the assessed value premium, since both groups are exposed to the same renovation-driven reassessment process.

The renovation-subsample hedonic estimate is 0.271 (SE = 0.127, $t = 2.13$, $p = 0.033$, implied 31.2 percent), reported in Table 1, column (3). The renovation-subsample DiD estimate is 0.380 (SE = 0.091, $t = 4.17$, $p < 0.001$, implied 46.3 percent), reported in column 4. Both estimates remain significant within the renovation-balanced subsample, increasing confidence that the premium reflects the market signal of green certification rather than renovation-driven reassessment.

IV. Permit-Anchored Event Study (Specification 3c)

Figure 9 presents the permit-anchored event study, which uses each parcel's first major renovation permit as the event anchor rather than the certification date, a choice necessary because uncertified control buildings never receive a certification date. The post-period extends to $k = +10$ to capture the average seven-year gap between permit issuance and certification in this sample. Pre-permit coefficients at $k \leq -3$ and $k = -2$ are near zero and statistically insignificant, meaning both groups were on the same trajectory before construction began, confirming parallel trends. Post-permit coefficients are near zero through $k = +3$ and grow significantly from $k = +4$ onward, reaching their largest values around $k = +7$ through $k \geq +10$, consistent with a premium that materializes upon certification rather than renovation alone.

V. Heterogeneity by Property Use Category

Figure 8 breaks down the hedonic certification premium by property use category. Hotels (COMH) show the largest coefficient at 1.251 (SE = 0.301, $p < 0.001$, implied 250 percent), though the treated sample is small ($N = 16$). Office buildings (COMO) yield a statistically

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significant premium of 0.330 (SE = 0.131, $p = 0.012$, implied 39 percent) across the largest treated group ($N = 227$). Mixed-commercial (COMM) estimates are large but marginally insignificant ($p = 0.055$). Retail and industrial premiums are positive but statistically indistinguishable from zero ($p = 0.141$ and 0.274), consistent with weaker incentives to signal energy performance in shorter-lease market segments.

VI. Conclusions

Taken together, the specifications tell a consistent story. A significant certification premium in assessed improvement value appears across all estimation approaches, ranging from 31 to 52 percent depending on the sample and method. The premium survives the renovation-matched comparison, the most demanding test, confirming that it reflects the market signal of green certification rather than concurrent capital investment. The permit-anchored event study further establishes that the premium materializes in the years following certification, not renovation, and grows over time. The heterogeneity results suggest the effect is concentrated in office and hotel properties, where ESG signaling and operational cost savings are most significant to tenants and investors. These patterns are consistent with the cost-side mechanism outlined in the conceptual model. Under San Francisco's predominantly gross lease structure, certification-related reductions in operating expenses raise NOI directly, and the SF Assessor's income-capitalization methodology translates that into higher assessed improvement values. These estimates may be biased in either direction depending on the period and the relative reassessment recency of treated and control buildings. Collectively, the findings support the hypothesis that green certification generates real, assessor-recognized value in San Francisco's commercial property market.

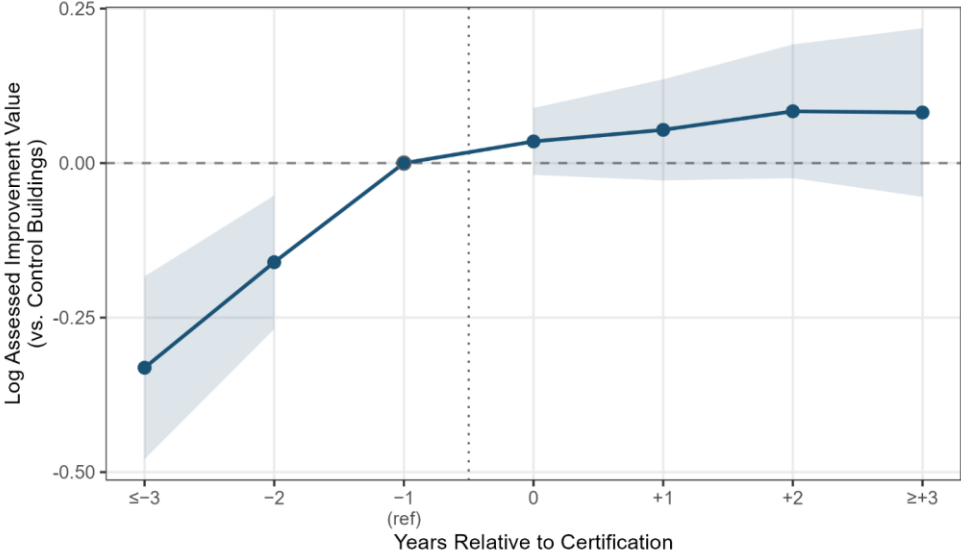
Commented [NSP7]: Maybe soften this a little as it is mentioned already elsewhere. Maybe add another paragraph to this to wrap it up.

Commented [NSP8]: I think having a table that shows all of the premium percentages could be super useful.

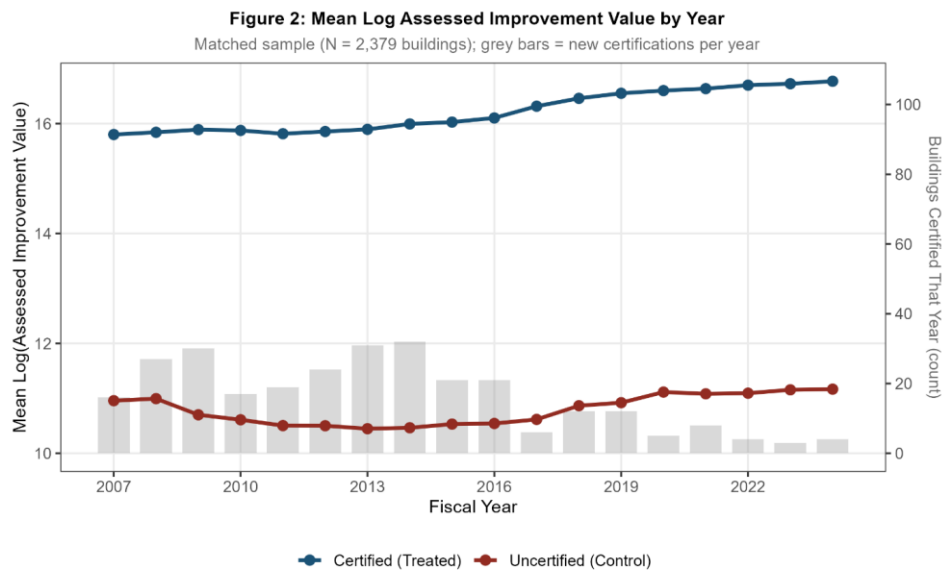
6. Appendix

Figure 1: Event Study - Assessed Improvement Value Relative to Certification Year

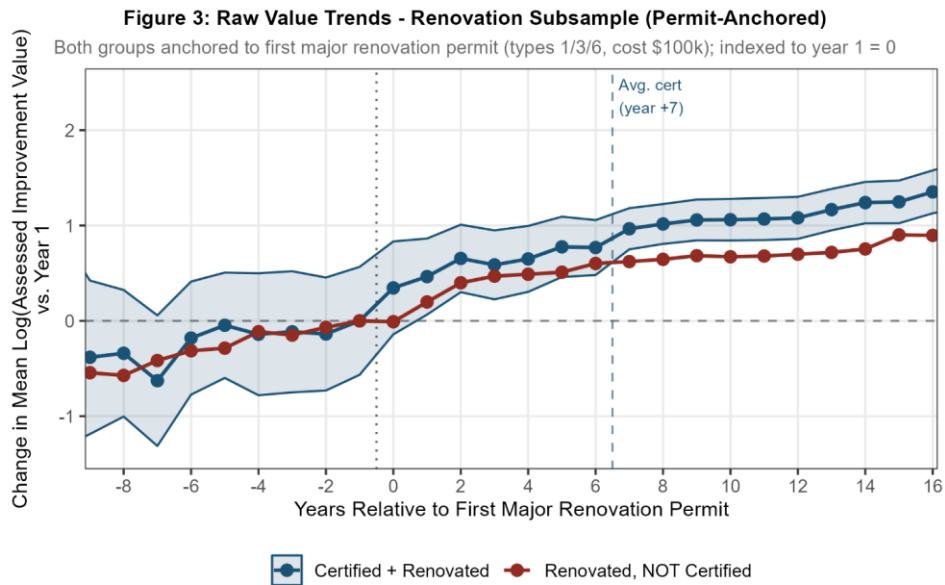
Two-way FE DiD panel; 95% CI shaded; reference period = year -1



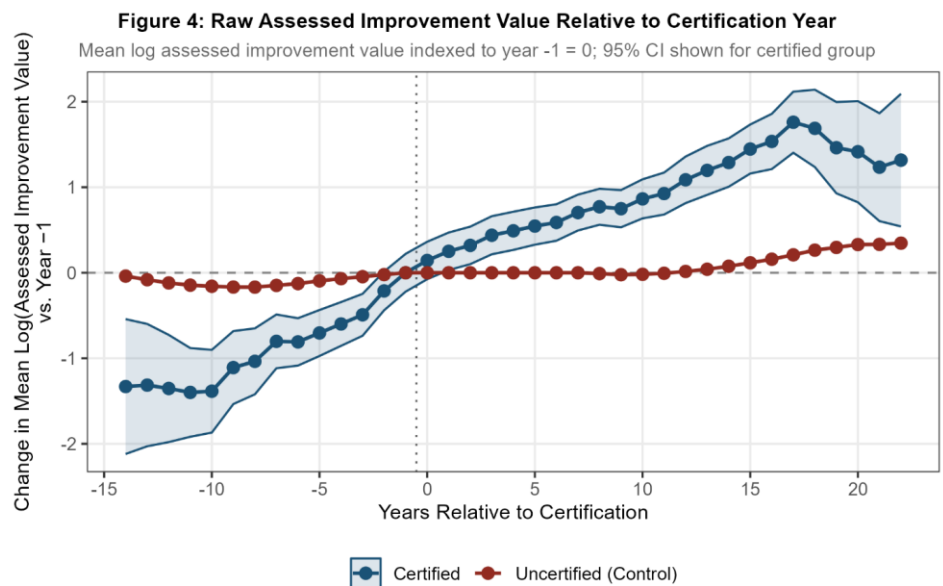
Event Study: All Matched Parcels. Plots the estimated log assessed improvement value for certified buildings relative to matched controls for each year from 3+ years before to 3+ years after certification (year -1 is the reference period, normalized to zero). Estimated using a two-way fixed effects DiD panel model with parcel-clustered standard errors. Used to assess whether pre-certification trends are parallel and when the post-certification value premium materializes.



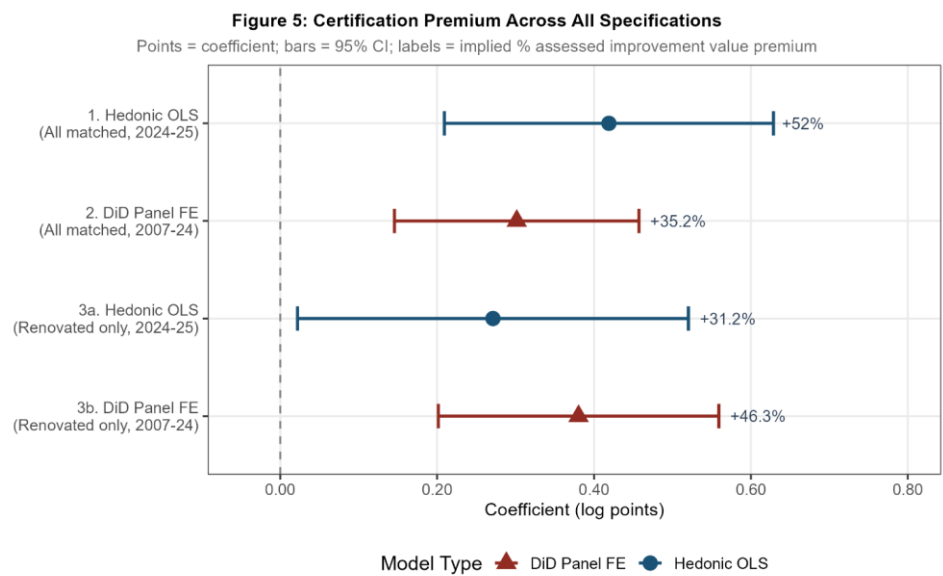
Mean Log Assessed Improvement Value by Year: Line chart of mean log assessed improvement value by year (2007–2024) for the treated and control groups separately. Provides an informal visual check of pre-trend parallelism and post-certification divergence between certified and matched uncertified buildings across the 18-year panel.



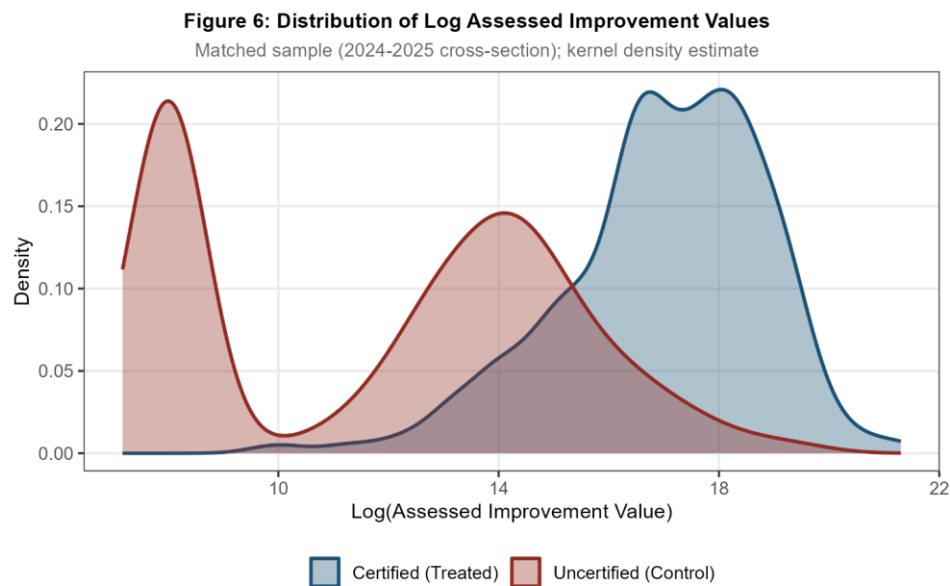
Raw Value Trends: Renovation Subsample. Restricted to buildings that had a major renovation permit (cost $\geq$ \$100,000). Compares value trajectories for certified + renovated buildings against renovated-but-uncertified buildings, allowing a visual check of whether renovation activity alone drives the pre-certification value dip observed in the full sample.



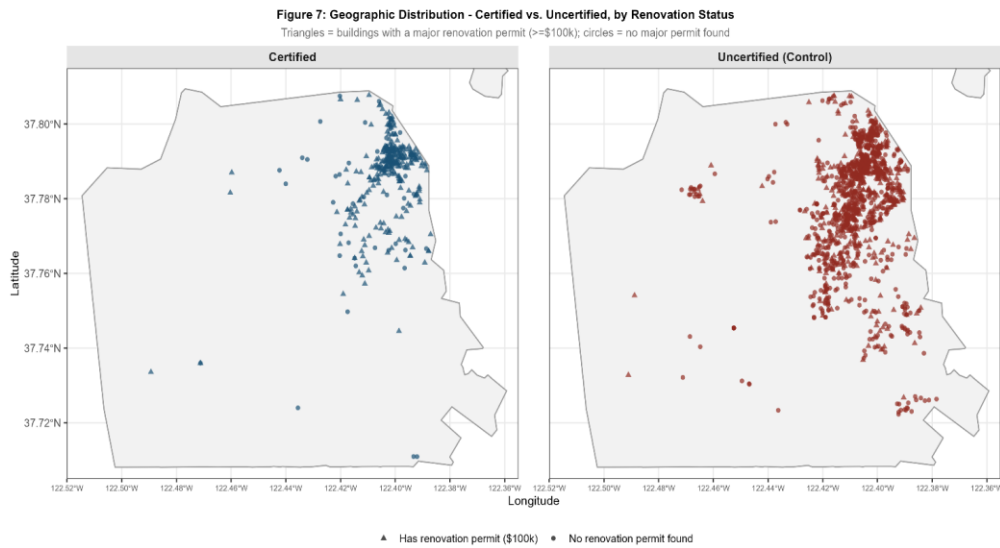
Raw Value Trends Relative to Certification Year: All Parcels. Plots mean log assessed improvement value indexed to each building's certification year (event time), separately for treated and control buildings. Complements the formal event study by showing the raw, unadjusted trajectory of assessed values around the certification event for the full matched sample.



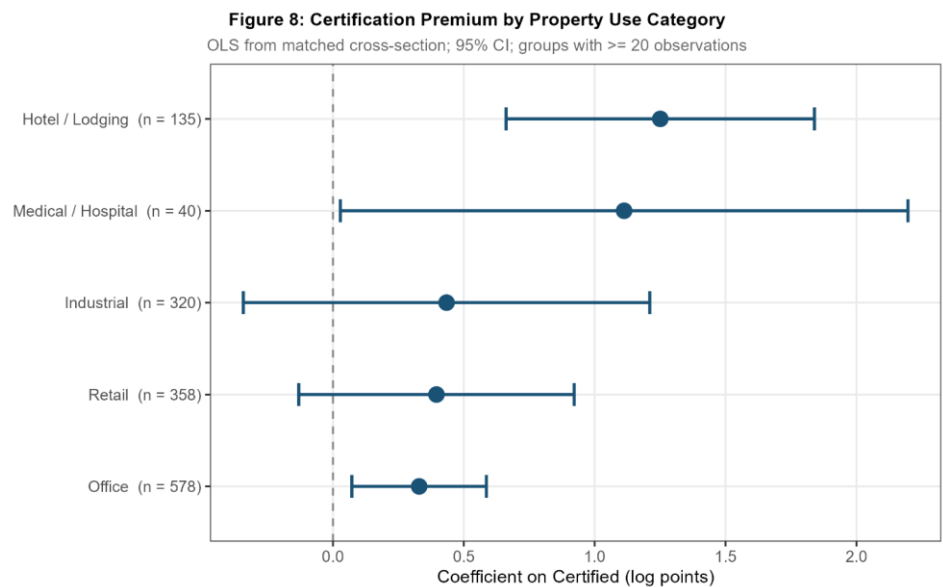
Forest Plot: Certification Premium Across All Specifications. Plots the β_1 point estimate and 95% confidence interval for each of the four main specifications (Regressions 1, 2, 3a, 3b) on a single axis, with the implied percentage premium labeled on each estimate. Allows direct visual comparison of the certification effect across hedonic cross-section and DiD panel models, and across the full matched sample versus the renovation subsample.



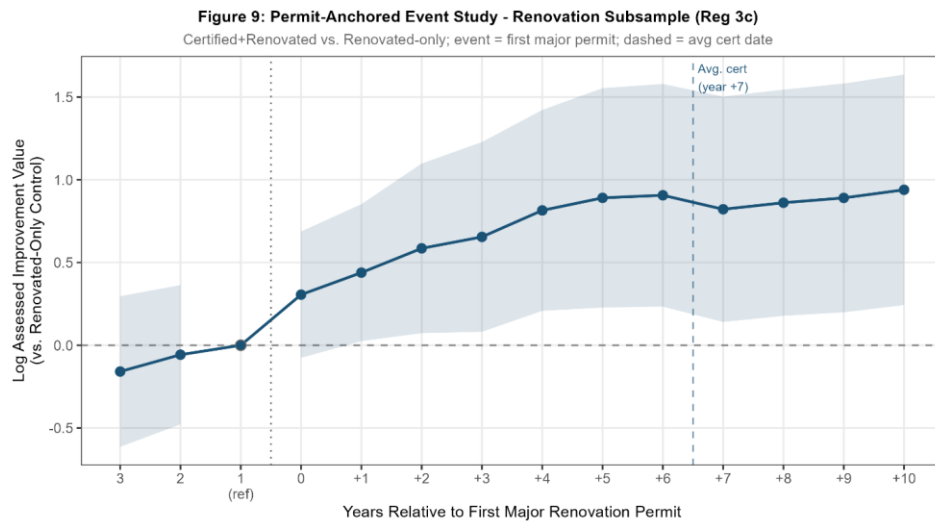
Distribution of Log Assessed Improvement Values by Certification Status. Kernel density curves of log assessed improvement value for certified versus uncertified parcels using the 2024–25 cross-section. Shows the full distributional difference between groups beyond the mean, illustrating that the premium is not driven solely by a small number of high-value outliers.



Geographic Distribution of Certified and Renovated Commercial Buildings. Map of San Francisco commercial parcels distinguished by both shape and color: shape indicates renovation permit status (circles = no renovation permit, triangles = renovation permit) and color indicates certification status. The spatial distribution of the study sample across SF's commercial districts and illustrates how certification and major renovation activity cluster geographically across commercial neighborhoods.



Certification Premium by Property Use Category. Coefficient plot showing how the estimated certification premium varies across property use categories (e.g., office, retail, mixed use). Identifies which building types drive the strongest and weakest certification effects, providing evidence on the heterogeneity of the green premium within the commercial sector.



Permit-Anchored Event Study: Renovation Subsample (Regression 3c). Event study using the first major renovation permit date as the event anchor rather than the certification date, with a longer post-period window of up to +10 years. Year +7 approximates the average gap between permit issuance and green certification. Provides an alternative identification test of whether assessed value gains follow certification or renovation activity.

Figure 10: Summary Statistics - SF Green Building Certification Study

PANEL A: SAMPLE BALANCE [mean (SD)]				
Variable	Certified (Treated)	Control	p-value (diff.)	
N parcels	346	2,070		
Log Assessed Improvement Value (2024-25)	16.98 (1.86)	11.87 (3.37)	<0.001***	
Square Footage (sq ft)	228740 (266481)	43184 (91916)	<0.001***	
Year Built	1952 (36)	1928 (27)	<0.001***	
% LEED Certified	65%	0%		
% EnergyStar Certified	76%	0%		
PANEL B: REGRESSION RESULTS				
Specification	Coef. (SE)	Implied Premium	95% CI	Sig.
1. Hedonic OLS (all matched, 2024-25)	0.419 (0.107)	+52.0%	[+23.3%, +87.5%]	***
2. DID Panel FE (all matched, 2007-24)	0.301 (0.080)	+35.2%	[+15.7%, +56.0%]	***
3a. Hedonic OLS (reno-only, 2024-25)	0.271 (0.127)	+31.2%	[+2.2%, +68.3%]	*
3b. DID Panel FE (reno-only, 2007-24)	0.380 (0.091)	+46.3%	[+22.3%, +74.9%]	***
3c. Permit-Anchored ES (reno-only, +7 yrs = avg cert)	0.822 (0.347)	+127.5%	[+15.1%, +349.4%]	*

Note: *** p<0.001 ** p<0.01 * p<0.05. Panel A parentheses = SD; Panel B parentheses = SE.

Summary Statistics Dashboard. Two-panel summary table. Panel A reports sample balance statistics, number of parcels, mean log assessed improvement value, square footage, year built, and share LEED or EnergyStar certified, for treated and control groups side by side. Panel B summarizes all six regression results, reporting β_1 , standard error, implied percentage premium, and significance level for each specification in a single consolidated table.